

HW3

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November 2025

1 Problem

A Radar Prototype, that helps the user see what distance is left to the object

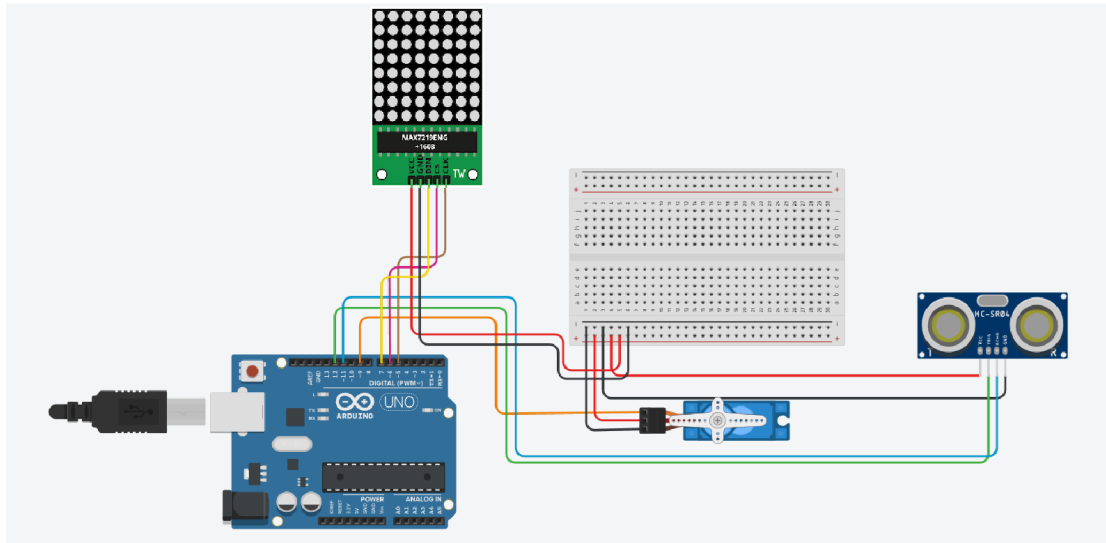
2 Design

The desing is simple: display, servo and ultrasonic sensor. The servo rotates 8 times at each direction (only 80 degrees, 10 degrees = 1 step). As it does it's one "step", the ultrasonic sensor triggers, send the information to the arduino, the arduino converts the data into human readable distance and displays it on the 8x8 LED matrica. rows - the 8 steps that the servo makes, columns - what is the distance between the ultrasonic sensor and the object (bottom - close, top - far)

3 parts

1x Arduino UNO R3
8x8 matrica with max7219 display driver
micro servo SG 90
ultrasonic sensor HC-SR04

4 wiring



5 What works /what didn't

the provided prototype works as it should be. What didn't work: not the best code, sometimes does a step and displays on the same row

6 Future improvements

fix the code, maybe a ping animation like on a sonar radar